

$$P = (2.5)^2 \times 2 = 12.5 \text{ W}$$

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B

The potential difference across the resistance R is given as $V = IR$.

Hence, $IR = E - Ir$

From the data, we form two equations:

$$3.0 = E - 1.0 r \dots (1)$$

$$4.8 = E - 0.4 r \dots (2)$$

Solving the equations simultaneously, $r = 3.0 \Omega$ and $E = 6.0 \text{ V}$

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A